MILITARY SPECIFICATION
CARTRIDGE, IMPULSE, XM259 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Impulse, XM259 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

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DRAWINGS

Munitions Command

D10551387	- Cartridge, Impulse, XM259, Assembly
F8797726	- Carton, Packing, Ammunition for Cartridge, Impulse, M67A1
F8797727	- Box, Packing, Ammunition for Cartridge, Impulse, M67A1
IEL-10551387	- Inspection Equipment List for Cartridge, Impulse, XM259

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, propellant, igniter or derangement of any component.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Ballistics.- The cartridge when tested in an acceptable XM26 Thruster mounted by the cap threads at the free end of a 45-pound simulated canopy and at an angle normal to the simulated canopy temperature conditioned at -65° Fahrenheit (F) to -70° F and at 200° to 205° F, shall rotate the 45-pound simulated canopy through an arc of 90 degrees and complete a stroke of 4.95" maximum. The average velocity of the simulated canopy during the thruster stroke shall be 33 feet per second (ft/sec) minimum. Ignition delay time shall not exceed 60 milliseconds (see 6.1). The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, in accordance with MS28741 at 70° ± 5° F.

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3.5 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity and other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 30 assembled cartridges for ballistics, 5 assembled cartridges for sealant application tests, 50 unplated cartridge cases for hardness tests, 23 sets of unassembled components with anodic finish for protective finish tests and 10 sets of unassembled components (including explosive components) for visual and dimensional examination and shall be tested as indicated in table I herein.

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Table I

Paragraphs Requirement	Paragraphs Test	Test	Sample Size	Type of Test Destr	Type of Test Nondestr
3.2	4.4.1	Protective Finish	23 ea Anodic Finish	X	-
3.3	4.4.2	Vibration	12 (from Ballistics)	-	X
Dwg	4.4.3	Hardness	50 ea	X	-
Dwg	4.4.4	Airtightness	ea	-	X
Dwg	4.4.5	Sealant Application	5 ea	X	-
3.4.2	4.4.6	Ballistics	30 ea	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with 3.4.2. The sample for firing shall consist of the following:

Table II

Temperature	Not Vibrated	Vibrated
-65° to -70° F	6	4
65° to 75° F	6	4
200° to 205° F	6	4

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product. - The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot. - Unless otherwise specified by the procuring agency, a lot shall consist of 300 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot
c. Primers from not more than one lot

4.3.2 Examination. - One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables III and IV. For an individual defect (single characteristic) the criteria in table III shall be used. For total defects within a major or minor group the criteria in table IV shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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Table IV

Total Allowable Defects per Sample Size

Number of Defects in Class	Sample Size for Major Defects			
	110	150	225	300
2 – 3	1	2	3	4
4 – 7	2	3	4	5
8 – 19	3	4	5	7
20 – 43	4	5	8	10
Over 43	6	8	11	14

Number of Defects in Class	Sample Size for Minor Defects			
	110	150	225	300
2 – 3	2	3	4	5
4 – 7	3	4	5	7
8 – 16	4	5	8	10
17 – 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Primer 72M, Drawing C7645332, covering a detail of Dwg D10551387.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Primer foil discolored	Visual
3. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.2 Head, Cartridge, Drawing C10521608, covering a detail of Dwg D10551387.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of primer pocket recess with primer pocket	Gage
102.	Concentricity of body with primer pocket	Gage
103.	Diameter of body, min	Gage
104.	Diameter of primer pocket	Gage
105.	Diameter of primer pocket recess, min	Gage
106.	Depth of primer pocket	Gage
107.	Thickness of base	Gage
108.	Metal defective	Visual
109.	Protective finish inadequate or defective	Visual
110.	Surface finish of primer pocket improper	Visual
111.	Burr	Visual
112.	Foreign matter	Visual
MINOR		
201.	Diameter of body, max	Gage
202.	Diameter of primer pocket recess, max	Gage
203.	Total length	Gage
204.	Angle on head	Gage
205.	Radius incorrect	Visual
206.	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Cartridge Case, Drawing C10551385, covering a detail of Dwg D10551387.

CRITICAL		
	None defined	
MAJOR		
101.	Concentricity of head with body	Gage
102.	Straightness of base	Gage
103.	Diameter of body, min	Gage

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	Defect	Method of Inspection
104.	Diameter of head, min	Gage
105.	Thickness of base, min	Gage
106.	Thickness of sidewall, min	Gage
107.	Metal defective	Visual
108.	Protective finish inadequate or defective	Visual
109.	Burr	Visual
110.	Foreign matter	Visual
MINOR		
201.	Diameter of body, max	Gage
202.	Diameter of head, max	Gage
203.	Total length	Gage
204.	Thickness of sidewall, max	Gage
205.	Location of datum diameter	Gage
206.	Radius or chamfer missing or incorrect	Visual
207.	Surface finish improper	Visual

4.3.2.4 Cartridge, Impulse, XM259, Assembly, Dwg D10551387.

CRITICAL		
1	Primer not properly sealed	Gage - X-ray 1/
2.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Concentricity of head with body	Gage
102.	Straightness of head	Gage
103.	Depth of recess on head	Gage
104.	Length of head	Gage
105.	Any part missing	X-ray/Visual 1/
106.	Improper assembly	X-ray/Visual 1/
107.	Crimp missing or improper	Visual
108.	Sealant missing	Visual
109.	Sealant in cavity	Visual
110.	Foreign matter	Visual
MINOR		
201.	Diameter, max	Gage
202.	Diameter of head, max	Gage
203.	Total length	Gage
204.	Location of datum diameter	Gage
205.	Varnish missing or inadequate	Visual

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1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Packing, Ammunition for Cartridge, Impulse, M67A1, Dwg F8797726.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Contents loose	Manual & Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damaged through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Ammunition for Cartridge, Impulse, M67A1, Dwg F8797727.

CRITICAL		
None defined		
MAJOR		
None defined		
MINOR		
201.	Contents loose	Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual

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Defect	Method of Inspection
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Protective finish.- Twenty-three units of each component of aluminum shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration.- Twelve of the 30 cartridges of the initial production sample shall be subjected to vibration. The 12 cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.3 Hardness.- Fifty units from each heat treated lot of components shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.5 Sealant application.- Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.- The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.- Thirty assembled cartridges from the initial production sample and 30 cartridges from each subsequent production lot shall be functionally tested, as prescribed in table V. Failure of any cartridge to comply with the requirement of 3.4.2 shall be cause for rejection of the lot.

Table V

Temperature	Sample Size
-65° to -70° F	10
65° to 75° F	10
200° to 205° F	10

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4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10551387.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Hardness.- A hardness testing machine shall be used for determining the hardness values specified on the applicable drawings.

4.4.4 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-10551387.

4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant is free of sealant.

4.4.6 Ballistics.- The cartridge shall be loaded into an acceptable XM26 Thruster divided into three equal groups and temperature conditioned in accordance with table V. The cartridge shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the thruster shall be reconditioned in accordance with 4.4.6.1.

4.4.6.1 Temperature conditioning.- An XM26 Thruster, loaded with a dummy XM259 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1-hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings F8797726 and F8797727.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ignition Delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

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